

CA2300: Do not use insecure deserializer BinaryFormatter (code analysis)

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Content

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CA2300: Do not use insecure deserializer BinaryFormatter

Summarize this article for me

Property	Value	Rule ID	CA2300	Title	Do not use insecure deserializer BinaryFormatter	Category	Security	Fix is breaking or non-breaking	Non-breaking	Enabled by default in .NET 10	No	Applicable languages	C# and Visual Basic
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Cause

A [System.Runtime.Serialization.Formatters.Binary.BinaryFormatter](#) deserialization method was called or referenced.

Rule description

Insecure deserializers are vulnerable when deserializing untrusted data. An attacker could modify the serialized data to include unexpected types to inject objects with malicious side effects. An attack against an insecure deserializer could, for example, execute commands on the underlying operating system, communicate over the network, or delete files.

This rule finds `System.Runtime.Serialization.Formatters.Binary.BinaryFormatter` deserialization method calls or references. If you want to deserialize only when the `Binder` property is set to restrict types, disable this rule and enable rules [CA2301](#) and [CA2302](#) instead. Limiting which types can be deserialized can help mitigate against known remote code execution attacks, but your deserialization will still be vulnerable to denial of service attacks.

`BinaryFormatter` is insecure and can't be made secure. For more information, see the [BinaryFormatter security guide](#).

How to fix violations

- Use a secure serializer instead, and **don't allow an attacker to specify an arbitrary type to deserialize**. For more information see the [Preferred alternatives](#).
- Make the serialized data tamper-proof. After serialization, cryptographically sign the serialized data. Before deserialization, validate the cryptographic signature. Protect the cryptographic key from being disclosed and design for key rotations.
- This option makes code vulnerable to denial of service attacks and possible remote code execution attacks in the future. For more information, see the [BinaryFormatter security guide](#). Restrict deserialized types. Implement a custom `System.Runtime.Serialization.SerializationBinder`. Before deserializing, set the `Binder` property to an instance of your custom `SerializationBinder` in all code paths. In the overridden `BindToType` method, if the type is unexpected, throw an exception to stop deserialization.

When to suppress warnings

`BinaryFormatter` is insecure and can't be made secure.

Pseudo-code examples

Violation

```
using System.IO;
using System.Runtime.Serialization.Formatters.Binary;

public class ExampleClass
{
    public object MyDeserialize(byte[] bytes)
    {
        BinaryFormatter formatter = new BinaryFormatter();
        return formatter.Deserialize(new MemoryStream(bytes));
    }
}
```

```
Imports System.IO
Imports System.Runtime.Serialization.Formatters.Binary

Public Class ExampleClass
    Public Function MyDeserialize(bytes As Byte()) As Object
        Dim formatter As BinaryFormatter = New BinaryFormatter()
        Return formatter.Deserialize(New MemoryStream(bytes))
    End Function
End Class
```

CA2301: Do not call `BinaryFormatter.Deserialize` without first setting `BinaryFormatter.Binder`

CA2302: Ensure `BinaryFormatter.Binder` is set before calling `BinaryFormatter.Deserialize`

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